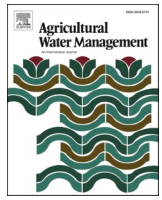
Agricultural Water Management 320 (2025) 109836



Contents lists available at ScienceDirect

Agricultural Water Management

journal homepage: www.elsevier.com/locate/agwat





Deep learning for intelligent irrigation decision-making: A review

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ARTICLE INFO

Keywords:
Deep learning
Irrigation decision-making
Water resource management
Precision agriculture

ABSTRACT

Global agriculture faces the dual challenges of water scarcity and climate change, making efficient and precise irrigation management increasingly important. This review analyzes the role of deep learning (DL) technologies in intelligent irrigation decision-making: (1) DL technologies have shifted irrigation management from experience-based decisions to data-driven precision prediction. (2) Deep learning architectures demonstrate distinct advantages in different aspects of irrigation management, including spatial identification, soil water content prediction, long-term forecasting, and optimization of water use. (3) Hybrid DL models often demonstrate superior performance in practical applications. (4) Edge-cloud collaborative architectures are particularly effective, reducing communication volume and decreasing response times from minutes to seconds. Despite progress, intelligent irrigation using DL faces challenges related to data quality, model generalization ability, and computational resource limitations, as well as application barriers such as cost, acceptance, and regional adaptability. Future work should prioritize climate-adaptive models, extreme-weather response, and ultra-precise management in water-scarce regions, while evaluating federated, few-shot learning and large language models as enabling methods.

1. Introduction

As the largest freshwater consumer, agriculture accounts for 70 % of global freshwater use (Kang et al., 2017; McDermid et al., 2023). Climate change and rising water demand are intensifying global water scarcity, a critical challenge for agriculture (Dinar et al., 2019; He et al., 2021; Qi et al., 2022). Approximately 3.8 million square kilometers of farmland worldwide are affected (Liu et al., 2022), and about 4 billion people (two-thirds of the global population) experience severe water scarcity for at least one month annually, with half residing in India and China (Mekonnen and Hoekstra, 2016). Low irrigation efficiency exacerbates water scarcity (McDermid et al., 2023). Developing efficient and precise irrigation decision support systems is essential.

Intelligent irrigation has transitioned from traditional experiential methods to Machine Learning (ML) and, more recently, to Deep Learning (DL) (Saggi and Jain, 2022; Umutoni and Samadi, 2024).

List of Abbreviations: AI, Artificial Intelligence; BCR, Benefit-Cost Ratio; BiLSTM, Bidirectional Long Short-Term Memory; CNN, Convolutional Neural Network; DL, Deep Learning; DNN, Deep Neural Network; DQN, Deep Q-Network; DRL, Deep Reinforcement Learning; ELM, Extreme Learning Machine; ET, Evapotranspiration; ETo, Reference Evapotranspiration; FAO, Food and Agriculture Organization; FNO, Fourier Neural Operator; GA, Genetic Algorithm; GAN, Generative Adversarial Network; GFS, Global Forecast System; GLDAS, Global Land Data Assimilation System; GRU, Gated Recurrent Unit; IoT, Internet of Things; IRR, Internal Rate of Return; IWP, Irrigation Water Productivity; LLM, Large Language Model; LoRa, Long Range; LSTM, Long Short-Term Memory; MAE, Mean Absolute Error; MAPE, Mean Absolute Percentage Error; ML, Machine Learning; MPC, Model Predictive Control; MSE, Mean Squared Error; NAS, Neural Architecture Search; NN, Neural Network; NPV, Net Present Value; NSE, Nash-Sutcliffe Efficiency; PCA, Principal Component Analysis; PINNs, Physics-Informed Neural Networks; PM, Penman-Monteith; PSO, Particle Swarm Optimization; PSPNet, Pyramid Scene Parsing Network; R^2, Coefficient of Determination; RE, Relative Error; ResBiLSTM, Residual Bidirectional Long Short-Term Memory; RF, Random Forest; RH, Relative Humidity; RL, Reinforcement Learning; RMSE, Root Mean Square Error; RNN, Recurrent Neural Network; RS, Remote Sensing; SEP, Standard Error of Prediction; SILO, Scientific Information for Land Owners; SVM, Support Vector Machine; SVR, Support Vector Regression; SWC, Soil Water Content; UAV, Unmanned Aerial Vehicle; U-Net, U-shaped Network; WS, Wind Speed; WUE, Water Use Efficiency.

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https://doi.org/10.1016/j.agwat.2025.109836
Received 10 May 2025; Received in revised form 15 September 2025; Accepted 19 September 2025
Available online 24 September 2025
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Traditional irrigation, reliant on experiential judgment or fixed scheduling, fails to address the complexities of modern irrigation decisions (Ara et al., 2021). Early ML applications mainly employed Support Vector Machines (SVM) (Gul et al., 2023), Random Forests (RF) (Mokhtar et al., 2022), and shallow Neural Networks (NNs) (Ahmed et al., 2023; Oymak and Soltanolkotabi, 2020), which performed well in processing structured data but had limitations in extracting deep features from complex spatiotemporal data. Advances in sensing, information, and communication technologies (ICT) have enabled DL to offer novel approaches for intelligent irrigation management (Saggi and Jain, 2022). Compared to traditional and ML methods, DL excels at processing massive heterogeneous data, automatically extracting features, and modeling complex non-linear relationships (Ahmed et al., 2023).

Table 1 contrasts traditional, ML, and DL approaches. While DL reduces manual feature design and models spatiotemporal patterns, it remains computationally intensive and less interpretable (Ahmed et al., 2023). The development of edge computing technology is gradually alleviating computational resource limitations. For instance, edge Artificial Intelligence (AI) systems that can detect water resource misuse events in real-time (Loukatos et al., 2022) and provide cloud-like functionality for IoT sensors (Ooi et al., 2023).

DL has become a key enabling technology for intelligent irrigation. Among different DL methods, convolutional Neural Networks (CNNs) excel in spatial data processing (Raei et al., 2022). The family of Recurrent Neural Networks (RNNs) excels in time series analysis (Jimenez et al., 2021). The Bidirectional Long Short-Term Memory (BiLSTM) model can predict Soil Water Content (SWC) with R^2 values of 0.818–0.991 (Yu et al., 2020). Deep Reinforcement Learning (DRL) is a promising approach for improving irrigation scheduling decisions, as demonstrated by the use of Deep Q-learning Networks (DQN) combined with weather forecasts, which achieved 23 mm water savings in Nanchang, China (Chen et al., 2021). Moreover, with the widespread application of the Internet of Things (IoT), the potential of DL at intelligent irrigation has been further demonstrated (Pang et al., 2023).

DL serves as a central hub for intelligent irrigation, integrating multi-source data to build predictive models and optimize decisions (Fig. 1). The data-driven process helps manage complex irrigation systems and balance agricultural production with sustainable water use (Cao et al., 2021).

This review strictly focuses on DL technologies in irrigation decision-making, which include water demand prediction, SWC monitoring, Evapotranspiration (ET) estimation, and irrigation scheduling optimization. We provide systematic guidance for researchers and practitioners to select appropriate technologies, identify research gaps, and foster innovation in DL applications for water resource management and sustainable agriculture. This review: (1) analyzed the application status and performance of DL technologies in various aspects of irrigation decision-making; (2) compared the advantages and disadvantages of

Table 1
Comparison of Traditional Irrigation, ML, and DL in Intelligent Irrigation.

Characteristic	Traditional Irrigation	Traditional ML	DL	Significance
Water Use Efficiency	Low efficiency, with significant water waste in conventional systems (Bwambale et al., 2022; Ma et al., 2022)	Improved efficiency, with documented water savings over traditional methods (Huang et al., 2023; Saggi and Jain, 2022; Tace et al., 2022)	Highest efficiency with 25–50 % water reduction compared to conventional methods (Addas et al., 2024; Ma et al., 2022; Pham et al., 2021)	Critical for sustainable water resource management in agriculture
Decision-making Approach	Rule-based decisions using fixed schedules or simple sensors (Bwambale et al., 2022; Kelly et al., 2024)	Data-driven predictions using structured data and pre-defined features (Goldstein et al., 2018; Tace et al., 2022)	End-to-end learning with automatic feature extraction from complex, multi-source data (Anuradha et al., 2024; Hashemi et al., 2024)	Impacts the flexibility and adaptability of irrigation management
Feature Engineering	Manual observations and measurements with limited variables (Arif et al., 2024; Saggi and Jain, 2022)	Often requires expert selection and transformation of relevant features (e.g., reducing from 26 to 11 inputs) (Goldstein et al., 2018; Sidhu et al., 2020)	Automates feature extraction from raw data; input variable selection remains crucial (Anuradha et al., 2024; El Sakka et al., 2025)	Reduces dependence on domain expertise and enables more complex modeling
Accuracy/ Precision	Low precision with over/under irrigation is common (Dinar et al., 2019; Umutoni and Samadi, 2024)	Medium-high precision (R^2 up to 0.91–0.99) for various water metrics (Gul et al., 2023; Liu et al., 2024; Tausif et al., 2023)	High precision, with models achieving R^2 up to 0.98 or higher (Tace et al., 2022; Yan et al., 2024; Yu et al., 2020)	Directly impacts crop yield, water conservation, and economic outcomes
Spatiotemporal Data Processing	Limited to point measurements at specific times (Arif et al., 2024; Chandrappa et al., 2023)	Handle some spatiotemporal relationships with explicit modeling (Nagaraj et al., 2021; Pagano et al., 2023)	Effectively captures complex spatial and temporal patterns (Chandrappa et al., 2023; Raei et al., 2022; Saraiva et al., 2020)	Enables comprehensive monitoring of field conditions across space and time
Climate Change Adaptability	Static systems with limited responsiveness to changing conditions (Elias et al., 2019; Liu et al., 2022)	Moderate adaptability through retraining with new data (Dang et al., 2024; Elias et al., 2023)	High adaptability with continuous learning for complex patterns (Chen et al., 2021; Shevchenko et al., 2024; Singh and Goyal, 2023)	Essential for agricultural resilience under increasing climate uncertainty
Data Integration Capability	Typically single-source data (e.g., manual readings) (Alfred et al., 2021; Saggi and Jain, 2022)	Integrate limited data types from multiple sources (Pagano et al., 2023; Yang et al., 2024)	Strong integration of heterogeneous data (satellite, sensor, weather) (Alfred et al., 2021; Cao et al., 2021; Wang et al., 2022)	Enables comprehensive monitoring and decision support
Real-time Response	Slow manual adjustments or simple automated timers (Bwambale et al., 2022; Loukatos et al., 2022)	Near real-time with moderate computational requirements (Bhoi et al., 2021; Sabrina et al., 2022)	Real-time with edge computing capabilities (Dahane et al., 2022; Loukatos et al., 2022; Ooi et al., 2023)	Critical for responding to rapidly changing field conditions
Implementation Complexity	Low technical complexity but labor-intensive (Saggi and Jain, 2022; Umutoni and Samadi, 2024)	Medium complexity requiring data collection and model development (Ara et al., 2021; Fünfgeld et al., 2019; Rodríguez Díaz et al., 2020)	High complexity with specialized expertise for development (Ahmed et al., 2023; Clay et al., 2024; Saggi and Jain, 2022)	Affects the adoption rate and practical deployment in agricultural settings
Cost Effectiveness	High operational costs for labor and water waste (Martey et al., 2023; Mendoza-Grimón et al., 2021)	Medium implementation costs with good return on investment (Christias and Mocanu, 2021; Goldstein et al., 2018)	High initial investment with the potential for significant long-term savings (Addas et al., 2024; Christias and Mocanu, 2021; Madhukumar et al., 2024)	Determines economic feasibility for different farm scales
Interpretability	Interpretable based on direct observations and simple rules (Ou et al., 2023)	Moderate to high interpretability depending on the algorithm (Ou et al., 2023; Paudel et al., 2023; Sabrina et al., 2022)	Often limited interpretability ("black box" nature), requiring specialized explanation methods (Kwaghtyo and Eke, 2023; Paudel et al., 2023)	Affects trust, troubleshooting, and regulatory compliance

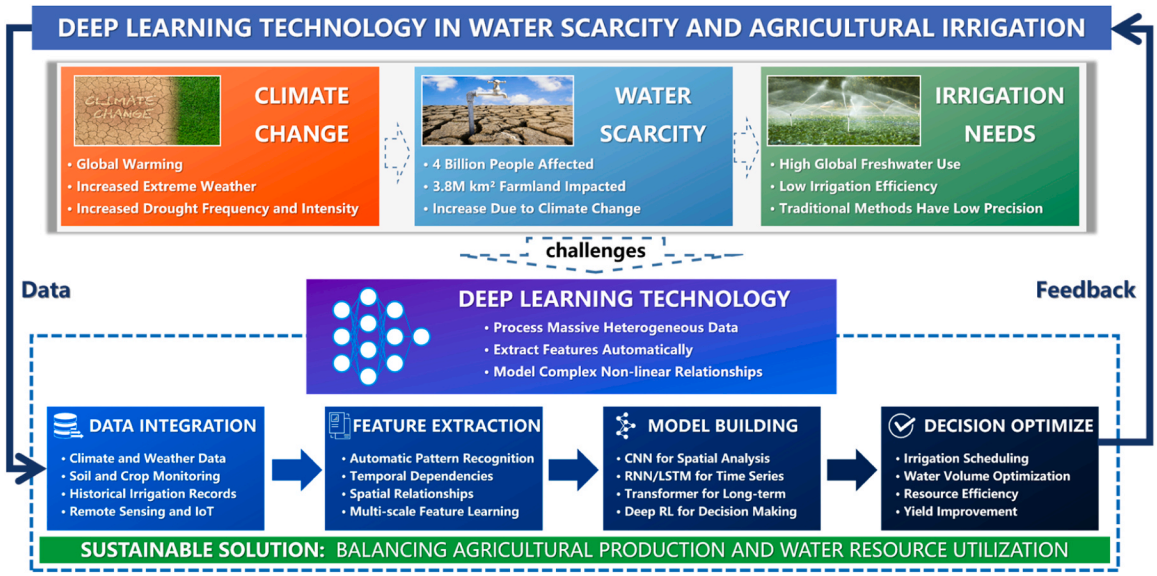


Fig. 1. Conceptual Framework of DL-Driven Intelligent Irrigation. The framework illustrates how DL technology addresses the challenges of climate change, water scarcity, and irrigation needs through a data-driven, systematic process to achieve sustainable solutions.

different DL methods in irrigation decision-making to guide technology selection; (3) summarized key challenges and corresponding solutions in the application process, including data acquisition, model training, and system deployment; (4) provided insights into future research directions and trends.

Following an overview of the theoretical foundations (Section 2), we dissect how DL transforms environmental sensing data (Section 3) into actionable water demand forecasts (Section 4). We then explore how DRL leverages these forecasts for optimal scheduling (Section 5). The review culminates by examining system architectures for real-world deployment (Section 6) and outlining the critical challenges and future frontiers for sustainable water management under climate change (Sections 7 and 8).

2. Theoretical foundation of DL in irrigation decision-making

2.1. Basic principles and architectures of DL

DL, a branch of ML using multi-level NNs (Ahmed et al., 2023), directly addresses irrigation challenges that traditional methods cannot solve. The primary advantage of DL is its capacity to process high-dimensional data, capturing nonlinear relationships, and enabling adaptive learning, which are essential for dynamic agricultural environments (Ahmed et al., 2023; Umutoni and Samadi, 2024). Different DL architectures are tailored for specific irrigation tasks (Fig. 2). CNNs excel in spatial analysis of remote sensing (RS) imagery (Younesi et al., 2024; Yu et al., 2021), while RNNs (LSTM/GRU) are adept at time-series predictions like SWC dynamics (Yu et al., 2020). Transformers are used for long-term forecasting by modeling long-range dependencies (Zheng et al., 2024), and DRL is for optimizing sequential decisions (Kelly et al., 2024). This section outlines the principles of these core architectures to guide their practical selection and application in irrigation.

2.1.1. Convolutional neural networks

CNNs are founded on the concepts of local receptive fields, weight sharing, and multi-layer feature extraction (Fig. 2a). CNNs achieve feature extraction through convolution operations, dimensionality reduction and invariance through pooling operations, while fully

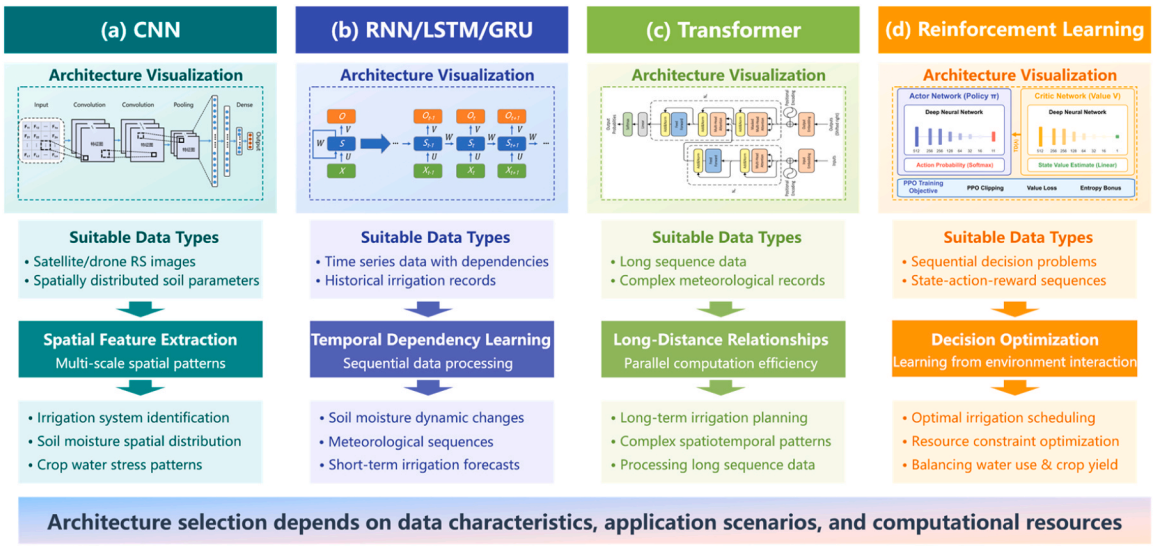


Fig. 2. Diagram Illustrating Major DL Architectures in Irrigation Decision-making.

connected layers integrate these features for final decision-making (Younesi et al., 2024). Formally:

$$H_l = f(W_l * H_{l-1} + b_l) \tag{1}$$

where H_l is the feature map of layer l, W_l is the convolution kernel, b_l is the bias, f is the activation function, and $*$ represents the convolution operation.

In intelligent irrigation, CNNs are suitable for processing data with spatial correlation, such as satellite and Unmanned Aerial Vehicle (UAV) RS images and spatially distributed soil parameters. The local receptive fields and hierarchical feature extraction of CNNs allow for the effective identification of the spatial structures of irrigation systems (Saraiva et al., 2020), detect spatial distributions of SWC (Singh and Gaurav, 2023), and analyze spatial patterns of crop water stress (Islam and Yamane, 2021). Hierarchical features make CNNs suitable for multi-scale RS tasks.

2.1.2. Recurrent neural networks and variants

RNNs rely on a recursive structure to process sequence data, maintain memory of past inputs through internal states (Landi et al., 2021) (Fig. 2b). The mathematical expression for a basic RNN unit is:

$$h_t = f(W_{xh}x_t + W_{hh}h_{t-1} + b_h) \tag{2}$$

$$y_t = g(W_{hy}h_t + b_y) \tag{3}$$

where h_t is the hidden state at time t, x_t is the input, y_t is the output, W is the weight matrix, b is the bias, and f and g are activation functions.

Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks address the vanishing gradient problem of traditional RNNs by introducing gate mechanisms (Al-Selwi et al., 2024). LSTM introduces input, forget, and output gates to control information flow (Landi et al., 2021), while GRU simplifies the LSTM structure with update and reset gates (Yu et al., 2021).

RNNs and their variants excel at processing data with temporal dependencies, such as SWC dynamic changes, meteorological time series data, and historical irrigation records. LSTM can effectively capture the long-term dependencies of irrigation-related time series (Jimenez et al., 2021), while GRU achieves a balance between computational efficiency and performance (Zheng et al., 2024). These models can incorporate past irrigation conditions and effects into current decision considerations, achieving adaptive adjustment.

2.1.3. Transformer architecture

The Transformer architecture, based on a self-attention mechanism, overcomes the sequential dependency limitations of RNNs (Madhukumar et al., 2024) (Fig. 2c). The mathematical expression for its core self-attention mechanism is:

$$\text{Attention}(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{4}$$

where Q, K, V are the query, key, and value matrices respectively, and d_k is the dimension of the key. Multi-head self-attention captures information from different representation subspaces by computing multiple sets of attention in parallel.

Transformers model long-range dependencies via self-attention, enabling long-horizon planning (Wang and Zha, 2024). Transformers can efficiently process long-sequence meteorological data and irrigation records (González Perea et al., 2024), achieving accurate predictions of future irrigation demands (Deforce et al., 2024). The parallel computation characteristics also make Transformers model training more efficient, suitable for processing large-scale irrigation data.

2.1.4. Deep reinforcement learning

DRL combines DL with Reinforcement Learning (RL), modeling the interaction between an agent and the environment through Markov decision processes (Kelly et al., 2024) (Fig. 2d). The core of DRL is the approximation of value functions or policy functions (Balderas et al., 2025), such as DQN approximating the Q-value function through NNs:

$$Q(s, a) \approx Q(s, a; \theta) \tag{5}$$

where s is the state, a is the action, and θ represents the NN parameters.

In intelligent irrigation decision optimization, DRL models the irrigation process as a sequential decision problem (Gautron et al., 2022). Irrigation decisions represent a policy optimization problem for an agent in a time-varying environment, where states include soil, crop, and meteorological conditions, actions are irrigation strategies, and rewards are Water Use Efficiency (WUE) or crop yield (Yang et al., 2020). DRL can learn optimal irrigation strategies in uncertain environments, handle complex constraint conditions, and adapt to dynamically changing environments (Chen et al., 2021).

2.2. Data characteristics and DL model selection

2.2.1. Analysis of irrigation decision data characteristics

Irrigation decision-making leverages heterogeneous data with distinct characteristics (Togneri et al., 2023). Meteorological data exhibit obvious temporal, periodic, and uncertain characteristics, with factors such as temperature, humidity, and precipitation directly affecting ET processes and crop water demands (Gul et al., 2023). Soil data exhibit spatial heterogeneity, time lag, and multi-scale characteristics, with complex spatiotemporal associations among soil parameters at different depths (Zheng et al., 2024), requiring models that can simultaneously process spatiotemporal relationships (Yu et al., 2020). RS images contain rich spatial features and multispectral information, existing as multi-dimensional tensors, suitable for large-scale irrigation area identification and crop condition monitoring (Saraiva et al., 2020).

Crop growth data exhibit nonlinear, stage-specific, and multi-factor correlation characteristics, with significant differences in water demands at different growth stages (Fuentes et al., 2019; Ramamoorthy et al., 2022). Water resource data, such as groundwater levels and reservoir levels, exhibit temporal, spatial correlation, and seasonal variation characteristics (Latif and Ahmed, 2023). Historical irrigation data are discrete event sequences with irregular time intervals and decision dependencies (Sidhu et al., 2020).

2.2.2. DL model selection theory

Effective DL model selection aligns data characteristics with model capabilities. For structured data types, CNNs are well-suited for grid-based data such as RS images by leveraging local receptive fields and weight sharing to extract spatial features (Saraiva et al., 2020), while RNNs and their variants are tailored to sequential data like meteorological time series through recurrent connections that capture temporal dependencies (Jimenez et al., 2021). For complex decision-making tasks such as irrigation scheduling, DRL provides a framework for policy optimization via agent–environment interactions (Yang et al., 2020). In resource-constrained contexts such as edge computing, parameter-efficient architectures (e.g., GRU over LSTM) are preferable to balance expressiveness and computational demand (Arif et al., 2024; Zheng et al., 2024), while real-time applications require models optimized for inference speed (Ooi et al., 2023).

Theoretically, multi-source heterogeneous data fusion can be approached through three strategies: early fusion, which combines raw inputs for homogeneous data; mid-fusion, which integrates feature-level representations for related heterogeneous sources; and late fusion, which aggregates decisions from independent models handling distinct data types (Cheng et al., 2023; Hashemi et al., 2024; Shuai et al., 2024; Singh and Gaurav, 2023). Table 2 summarizes the data–model alignment for irrigation decisions. Model selection should be based on data structure matching principles, achieving a balance between accurately

Table 2
Irrigation Decision Data Types and Suitable DL Models.

Data Type	Data Characteristics	Suitable DL Models	Theoretical Applicability Basis
Meteorological Data	Temporality, periodicity, uncertainty (Gul et al., 2023; Nagappan et al., 2020)	LSTM (Latif and Ahmed, 2023), GRU (Gao et al., 2025), Transformer (Wang and Zha, 2024)	Effectively processes temporal dependencies and long-term patterns (Gao et al., 2025; Nagappan et al., 2020)
Soil Data	Spatial heterogeneity, time lag, multi-scale (Aranda Britez et al., 2025; Han et al., 2024; Zheng et al., 2024)	CNN (Wu et al., 2025), LSTM (Li et al., 2022), GRU (Zheng et al., 2024), Transformer (Wang and Zha, 2024)	Simultaneously processes spatial distribution features and temporal evolution laws (Wu et al., 2025; Zheng et al., 2024)
RS Images	Spatial features, spatiotemporal changes, multispectral (Raei et al., 2022; Saraiva et al., 2020)	CNN (Saraiva et al., 2020), U-shaped Network (U-Net) (Raei et al., 2022), Pyramid Scene Parsing Network (PSPNet) (Zhang et al., 2020)	Excels at extracting multi-scale spatial features and pixel-level classification (Paolini et al., 2022; Zhang et al., 2020)
Crop Growth Data	Nonlinearity, stage specificity, multi-factor correlation (Fuentes et al., 2019)	Deep Neural Network (DNN) (Fuentes et al., 2019), CNN (Okayama et al., 2025), LSTM (Paudel et al., 2023)	Captures complex nonlinear relationships and temporal development patterns (Paudel et al., 2023; Ramamoorthy et al., 2022)
Water Resource Data	Temporality, spatial correlation, seasonality (He et al., 2023; Latif and Ahmed, 2023)	LSTM (Latif and Ahmed, 2023), GRU (He et al., 2023), Transformer (Deforce et al., 2024)	Processes long-term dependencies and periodic patterns (Dang et al., 2024; Latif and Ahmed, 2023)
Historical Irrigation Data	Discrete events, irregular time intervals (Kelly et al., 2024; Yan et al., 2024)	LSTM (Yan et al., 2024), GRU (Ahmed et al., 2021), RL (Balderas et al., 2025; Kelly et al., 2024)	Processes irregular time series and decision optimization problems (Kelly et al., 2024)

expressing data characteristics and computational efficiency.

2.2.3. *Heterogeneous data fusion for irrigation decision-making*

Intelligent irrigation decision-making requires integrating heterogeneous data, for which DL provides two main theoretical frameworks for fusion: feature-level and decision-level.

Feature-level fusion maps different data types into a shared feature space (Huang et al., 2024). This includes using Multimodal Encoder-Decoders to find cross-modal relationships (Han et al., 2024; Li et al., 2022), Tensor Decomposition for multi-dimensional data structures (Basargin et al., 2023), and Graph Neural Networks to model complex variable dependencies in irrigation systems (Bai and Tahmasebi, 2023).

Alternatively, decision-level fusion uses ensemble learning to combine the outputs of multiple models, thereby improving overall performance (Shuai et al., 2024). Common methods are voting to reduce prediction bias (Cheng et al., 2023), stacked generalization for a more flexible fusion mechanism (Ibrahim et al., 2024), and Bayesian Model Averaging to account for prediction uncertainty in risk assessment (Gao et al., 2024).

Selecting an appropriate fusion strategy requires analysis of data characteristics and task requirements. In practice, hybrid architectures such as CNN-GRU (Yu et al., 2021) and GRU-Transformer (Zheng et al., 2024) often process complex data in irrigation prediction more effectively.

2.3. *Evaluation metrics and performance analysis methods*

Predictive models are evaluated across accuracy, consistency, and stability dimensions, measuring error rates, trend capture capability, and performance under varying conditions. Decision optimization models are assessed based on resource benefits (e.g., WUE), economic returns, and environmental impacts. Fair model comparison requires controlling data equivalence, standardizing computational resources, and maintaining consistent hyperparameter optimization approaches within a multi-dimensional evaluation framework. These metrics provide essential guidance for selecting, improving, and implementing DL technologies that promote efficient water resource utilization and sustainable agricultural development (Table 3).

2.4. *Discussion*

The core theoretical challenge in irrigation decision-making modeling involves balancing the high expressivity of data-driven models with the non-negotiable physical constraints of hydrological systems. Purely data-driven models, while effective in data-rich settings, may violate hydrological principles. Conversely, imposing overly strict physical constraints can hinder a model's ability to capture complex nonlinearities. Physics-Informed Neural Networks (PINNs), incorporate physical laws like water conservation into the loss function to integrate domain knowledge (Haruzi and Moreno, 2023). Neural operators like DeepONet (Taccari et al., 2024) and the Fourier Neural Operator (FNO) (Pang, 2024) learn solution operators of physical equations, maintaining flexibility while adhering to physical plausibility (Li et al., 2024).

A further challenge is cross-scale consistency. Integrating microscale data (e.g., SWC) with macroscale considerations (e.g., regional water allocation) in irrigation decisions presents challenges in uncertainty propagation and information loss across scales. Although multi-scale architectures can process different resolutions in parallel, ensuring cross-scale consistency and physical validity remains a key research challenge. Future work should transition from fragmented "prediction-then-decision" models to end-to-end learning systems that directly map environmental observations to irrigation actions, while incorporating physical constraints and quantifying uncertainties. DRL offers a promising framework, however, developing robust agents capable of handling these compound uncertainties remains a critical research frontier. Advances in Bayesian DL present a viable path toward addressing these challenges (Madhukumar et al., 2024).

3. DL in environmental sensing and monitoring

3.1. *Spatial identification and classification of irrigation systems*

U-Net architectures with ResNet-34 backbones achieve 70–86 % accuracy in complex irrigation scene recognition (Raei et al., 2022). Enhanced U-Net implementations further improve performance for center pivot irrigation systems (99 % precision, 88 % recall) (Saraiva et al., 2020). These CNN-based approaches excel by capturing multi-scale spatial features and spatial heterogeneity simultaneously.

The global irrigation mapping work by Nagaraj et al. (2021) demonstrates scalability to continental levels at 9-kilometer resolution. Further demonstrating high precision, a study using a time series of RS data and a ResNet model achieved a final accuracy of 90.1 % for classifying different irrigation systems at the field scale (Paolini et al., 2022).

Key challenges remain in computational efficiency and

Table 3
Evaluation Metrics for Predictive and Decision Optimization Models in DL-Based Irrigation.

Model Type	Evaluation Dimension	Evaluation Metric	Calculation Formula	Rationale and Application
Predictive Models	Accuracy Evaluation	Root Mean Square Error (RMSE)	$\sqrt{\frac{1}{n}\sum_{i=1}^{n}(P_i - O_i)^2}$	Sensitive to large errors; evaluates ability to avoid serious prediction errors (Yu et al., 2020)
		Mean Absolute Error (MAE)	$\frac{1}{n}\sum_{i=1}^{n}\lvert P_i - O_i\rvert$	Treats all errors equally; evaluates overall stability (Datta and Faroughi, 2023)
		Relative Error (RE)	$\frac{\lvert P_i - O_i\rvert}{O_i}$	Compares accuracy at different scales; suitable for cross-regional comparison (He et al., 2023)
	Consistency Evaluation	Coefficient of Determination (R^2)	$1 - \frac{\sum_{i=1}^{n}(O_i - P_i)^2}{\sum_{i=1}^{n}(O_i - \overline{O})^2}$	Measures explained variation; evaluates trend capture ability (Gul et al., 2023)
		Nash-Sutcliffe Efficiency (NSE)		Widely used in hydrology; evaluates reliability of predictions (Band et al., 2020)
	Stability Evaluation	Cross Validation	e.g., K-fold validation	Evaluates generalization ability across different data subsets (Shiri et al., 2020)
		Sensitivity Analysis	e.g., One-at-a-time analysis	Tests model response to input changes and noise (Singh and Gaurav, 2023)
		Uncertainty Quantification	e.g., Prediction interval coverage probability	Provides risk assessment through probabilistic analysis (Ara et al., 2021)
Decision Optimization Models	Resource Benefits	WUE	$\frac{Y}{W_I}$	Measures output efficiency per unit water (Mendoza-Grimón et al., 2021)
		Irrigation Water Productivity (IWP)	$\frac{E_R}{W_I}$	Evaluates economic return on irrigation water (Temnani et al., 2022)
		Resource Saving Rate	$\frac{C_B - C_O}{C_B}$	Quantifies resource savings from optimization (Chen et al., 2021)
	Economic Benefits	Net Present Value (NPV)	$\sum_{t=0}^{n}\frac{B_t - C_t}{(1+r)^t}$	Considers the time value of investments and returns (Yang et al., 2020)
		Benefit-Cost Ratio (BCR)	$\frac{\sum_{t=0}^{n}\frac{B_t}{(1+r)^t}}{\sum_{t=0}^{n}\frac{C_t}{(1+r)^t}}$	Evaluates investment efficiency (Christias and Mocanu, 2021)
		Internal Rate of Return (IRR)	Value of r that makes $\sum_{t=0}^{n}\frac{B_t - C_t}{(1+r)^t} = 0$	Evaluates investment rate of return (Christias and Mocanu, 2021)
	Environmental Benefits	Energy Efficiency	$\frac{B_I}{E_C}$	Evaluates energy utilization efficiency (Mohammed et al., 2023)
		Carbon Footprint Reduction	$\frac{E_B - E_O}{E_B}$	Quantifies climate impact improvement (Ochoa-Barragán et al., 2024)
		Ecosystem Service Value	Varies by method	Monetary evaluation of biodiversity and ecosystem impacts (Anuradha et al., 2024)

Note: In the formulas above: P_i = predicted value; O_i = observed value; $\overline{O}$ = mean of observed values; n = number of samples; Y = crop yield; W_I = irrigation water volume; E_R = economic return; C_B = baseline consumption; C_O = optimized consumption; B_t = benefits in period t; C_t = costs in period t; r = discount rate; B_I = irrigation benefit; E_C = energy consumption; E_B = baseline carbon emissions; E_O = optimized carbon emissions.

generalizability across diverse landscapes. Lightweight CNN variants and specialized agricultural RS architectures represent critical future development directions (El Sakka et al., 2025; Jiang et al., 2021; Xiao et al., 2024).

3.2. Multi-depth SWC monitoring and estimation

DL mitigates the challenges of SWC monitoring, spatial heterogeneity, temporal dynamics, and sensor deployment costs, by enhancing measurement precision and providing innovative estimation approaches.

CNN and DNN calibration improved readings for 84.83 % of sensors, providing a more agile and cost-effective implementation than traditional approaches (Aranda Britez et al., 2025). For field-scale assessment from limited point measurements, DL-based approaches integrate ground sensors with satellite data to generate high-resolution moisture maps (R = 0.80) across diverse landscapes(Singh and Gaurav, 2023).

The hierarchical approach developed by Chandrappa et al. (2023) achieves 90 % success in predicting deep-layer moisture states using only shallow-layer data, reducing sensor requirements. Rodić et al. (2022) employs a Long Range (LoRa)-based system that eliminates traditional sensors, instead using signal analysis from underground beacons to sense moisture with minimal power consumption and extended battery life.

Multi-source fusion strategies enhance monitoring capabilities. The multimodal approach by Zhang et al. (2023) integrates thermal infrared and multispectral data, accounting for vegetation stage effects on measurement accuracy. Yu et al. (2022) demonstrated that integrating point measurements with satellite-derived indices reduces RMSE by 15.1 % compared to GIS-based spatial interpolation methods.

3.3. DL approaches for evapotranspiration monitoring

Accurate ET estimation is crucial for irrigation decision-making, and different DL architectures offer distinct advantages in this field (Table 4). A DNN combined with Principal Component Analysis (PCA) can reduce the required meteorological variables, lowering computational needs while maintaining high accuracy, which is particularly suitable for data-limited areas (Nagappan et al., 2020). Hybrid architectures often demonstrate superior performance. CNN-XGB and CNN-SVR models achieve high accuracy in rice and wheat ET prediction (NSE values of 0.95 and 0.97) and can predict future increases in crop water consumption, aiding long-term water resource planning (Sharma et al., 2022). For extremely data-scarce regions, a PSO-ELM model can achieve superior prediction using only temperature data, providing a practical solution for resource-poor areas (Zhu et al., 2020).

However, DL is not the optimal choice for all scenarios. An optimized SVM model (R^2 = 0.9970) was found to slightly outperform a DL-LSTM model (R^2 = 0.9688), underscoring the importance of selecting models based on specific data characteristics and application needs (Gul et al.,

Table 4
Performance of DL Architectures in Evapotranspiration Estimation Scenarios.

Model	Reference	Application	Inputs	Performance	Contribution
DNN	(Nagappan et al., 2020)	Wetland	3 principal components (via PCA)	Outperforms FAO-56 PM	Higher accuracy reduces computational complexity
CNN-XGB	(Sharma et al., 2022)	Rice	Climate data	NSE = 0.95	Adapts to future climate change
CNN-SVR	(Sharma et al., 2022)	Wheat		NSE = 0.97	
GA-ELM	(Kiraga et al., 2024)	Alfalfa	Meteorological data	Outperforms other ML models	Suitable for various time scales
PSO-ELM	(Zhu et al., 2020)	Gobi Desert	Temperature data only	Outperforms six empirical models	Usable in extremely limited data scenarios
RF Bagging	(Tausif et al., 2023)	Arid farmland	Minimum meteorological parameters (Tmin, Tmax, RH, WS)	f-measure = 93.15 %	RMSE and MAE were reduced by 17 % and 10 %
LSTM	(Dang et al., 2024)	Under climate change (Jinghiqu Irrigation Area, China)	Meteorological data	NSE = 0.86, r = 0.93	Adapts to future climate change
SVM vs. LSTM	(Gul et al., 2023)	Semiarid agricultural region (Pakistan)	Meteorological data	R^2 = 0.9970 (SVM)/0.9688 (LSTM)	DL is not universally superior and model selection is critical.
GRU	(Gao et al., 2025)	Yulin City, China	Vapor pressure, solar radiation	Error range −1.5–1.5 mm/day (7-step ahead)	Superior multi-step forecasting with quantum-inspired attention mechanism

2023). This study also revealed that the best variable combination can improve prediction accuracy by 15–20 %, a key consideration for developing lightweight models for edge computing environments.

3.4. Edge computing for real-time SWC monitoring

Edge computing reduces latency, bandwidth use, and reliability issues in rural monitoring. The lightweight edge AI system achieves real-time anomaly detection with minimal resources, reducing communication volume through event-driven strategies and decreasing response time from minutes to seconds (Loukatos et al., 2022). Edge-cloud collaborative architectures optimize task distribution, deploying lightweight models at the edge while reserving complex training for cloud infrastructure (Ooi et al., 2023).

Sensor network optimization through DL reduces hardware requirements while maintaining coverage. Virtual sensing frameworks estimate SWC at unmonitored locations, extending coverage without additional hardware (Patrizi et al., 2022). Real-time anomaly detection frameworks (autoencoder and GAN) distinguish between SWC sensor malfunctions and actual irrigation issues, critical for maintaining system reliability and user trust (Sabrina et al., 2022).

3.5. Discussion

Signal-based LoRa systems that function with minimal power and hierarchical approaches in deep soil prediction from shallow measurements demonstrate that DL can reduce physical sensor requirements while expanding monitoring coverage (Chandrappa et al., 2023; Rodić et al., 2022). This transition is enabled by the capacity of CNNs to extract spatial patterns from RS data and of RNNs/LSTMs to model temporal soil-water dynamics, creating a computational foundation for virtual sensing, where model-based inference effectively reduces sensor density.

By exploiting environmental data's inherent structure: spatial continuity, temporal patterns, and cross-modal correlations, DL effectively reduces the dimensionality of the sensing problem. Edge-cloud collaborative architectures transform these theoretical advantages into practical systems by deploying lightweight models at field edges.

4. DL in water demand prediction

Irrigation water requirement, a key metric derived from ET and adjusted for factors like effective rainfall, is the amount of water that must be artificially applied to meet crop needs. This chapter describes how DL predicts future irrigation water requirements across multiple spatial and temporal scales.

4.1. Spatial prediction and mapping of irrigation water requirements

Spatial prediction generates high-resolution maps of water requirements across agricultural landscapes, supports regional planning and precision irrigation (El Sakka et al., 2025). The CNN framework provides end-to-end predictions, processing multi-source RS data to generate irrigation demand forecasts (Fig. 3). Zhang et al. (2023) propose UAV-based multi-modal sensing combined with ML algorithms, which achieves accurate SWC estimations across crop types, growth stages, and irrigation treatments (R^2 = 0.68–0.78, RMSE = 19.36–20.82 %), highlighting the potential of ML methods for supporting data-driven irrigation planning. Quantitative prediction at the county-level demonstrates high precision (R^2>0.97, RMSE=0.06 km^3) through Gaussian process regression integration with hierarchical sample weighting to address data imbalance (Emami et al., 2022). Multi-source integration frameworks fuse satellite imagery with meteorological data and soil characteristics, improving SWC prediction accuracy ($R^2 \in [0.32, 0.52]$) over single-source approaches ($R^2 \in [0.11, 0.19]$) (Yang et al., 2024). Multi-source integration enables robust predictions across diverse landscapes with varying sensor coverage.

4.2. Temporal irrigation demand forecast with RNN

Temporal forecasting estimates future water demand from historical and current conditions. RNNs, particularly the LSTM and GRU variants, are commonly used for temporal prediction (Docheshmeh Gorgij et al., 2023; He et al., 2023). The gated memory mechanisms of these models effectively capture seasonal patterns and long-term dependencies of irrigation demands, and adapts to crop phenology and weather variability (He et al., 2023). LSTM models demonstrate superior short-term prediction capabilities (Jimenez et al., 2021), achieving R^2= 0.82–0.98 for hour-ahead predictions. GRU variants (Gao et al., 2025; Zheng et al., 2024) balance accuracy with computational efficiency for medium-term horizons, while bidirectional architectures excel in season-long predictions. By processing information in both forward and backward directions, these models capture dependencies from both past and future contexts (Yu et al., 2020).

Advanced temporal models enhance prediction stability under variable conditions. Multi-head LSTM approaches use specialized prediction heads for different time horizons, achieving consistent performance (R^2>0.95) across short to long-term predictions (Datta and Faroughi,

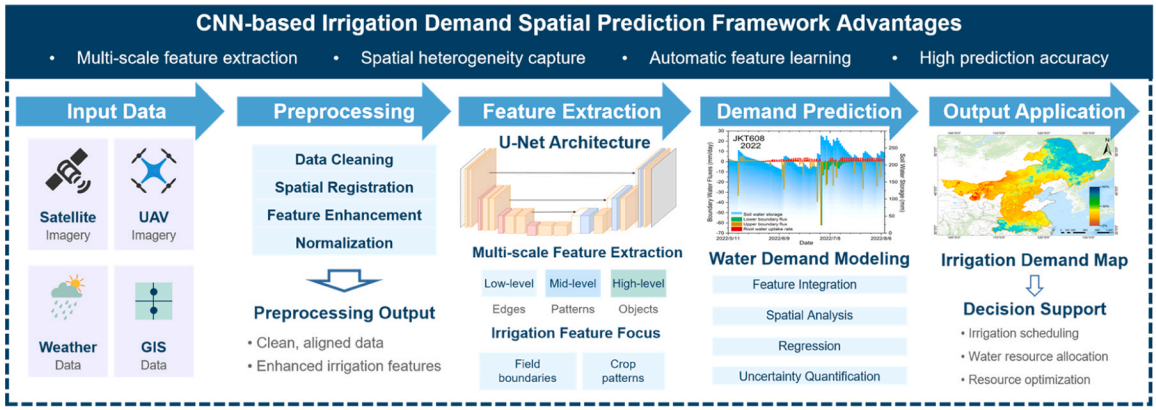


Fig. 3. CNN-based Irrigation Demand Spatial Prediction Framework.

2023). Causal models explicitly represent causal relationships between environmental factors and irrigation demands. The R^2 improvements for the proposed Causal-ConvLSTM over ConvLSTM are 10.17 %, 18.40 %, and 25.65 % at 1, 7, and 14 lead times, respectively (Wu et al., 2025). The primary challenges for temporal models include cumulative errors in extended sequences and limited adaptability to anomalous events. These issues are particularly pronounced in agricultural applications where extreme weather events impact irrigation demands.

4.3. Long-term irrigation planning with transformer

Transformer architectures excel at long-term planning, as they capture extended temporal dependencies while accounting for seasonal and climatic variability (Deforce et al., 2024). Unlike sequential RNN processing, Transformers simultaneously consider all sequence elements, directly modeling associations between arbitrary time points and more effectively capturing long-term dependencies (Fig. 4).

Transformer architectures excel at long-term planning by directly modeling extended temporal dependencies to account for seasonal and climatic variability (Fig. 4) (Deforce et al., 2024). A computationally efficient model using Transformers for week-ahead prediction improved accuracy by 89.8 % over traditional methods, forecasting 99.9 % of irrigation scenarios with average errors of 2.10 % (González Perea et al., 2024). The multi-head attention mechanism simultaneously learns different temporal representations, capturing both immediate fluctuations and long-term trends required for seasonal planning under climate change (Gao et al., 2025). Advanced implementations address key limitations, with sparse attention mechanisms reducing computational requirements and uncertainty-aware variants providing probabilistic predictions for risk assessment (Dong et al., 2024; Wang et al., 2024).

4.4. Hybrid models for enhanced prediction performance

Hybrid DL architectures overcome the limitations of single models by integrating the complementary strengths of different approaches. In irrigation, these architectures are particularly suitable for processing spatiotemporal composite data, where CNNs extract spatial features, RNNs process temporal dependencies, and Transformers capture long-distance relationships. Table 5 compares the performance characteristics of different hybrid architectures, revealing their differentiated advantages in irrigation prediction.

Hybrid DL architectures derive their theoretical advantage from a

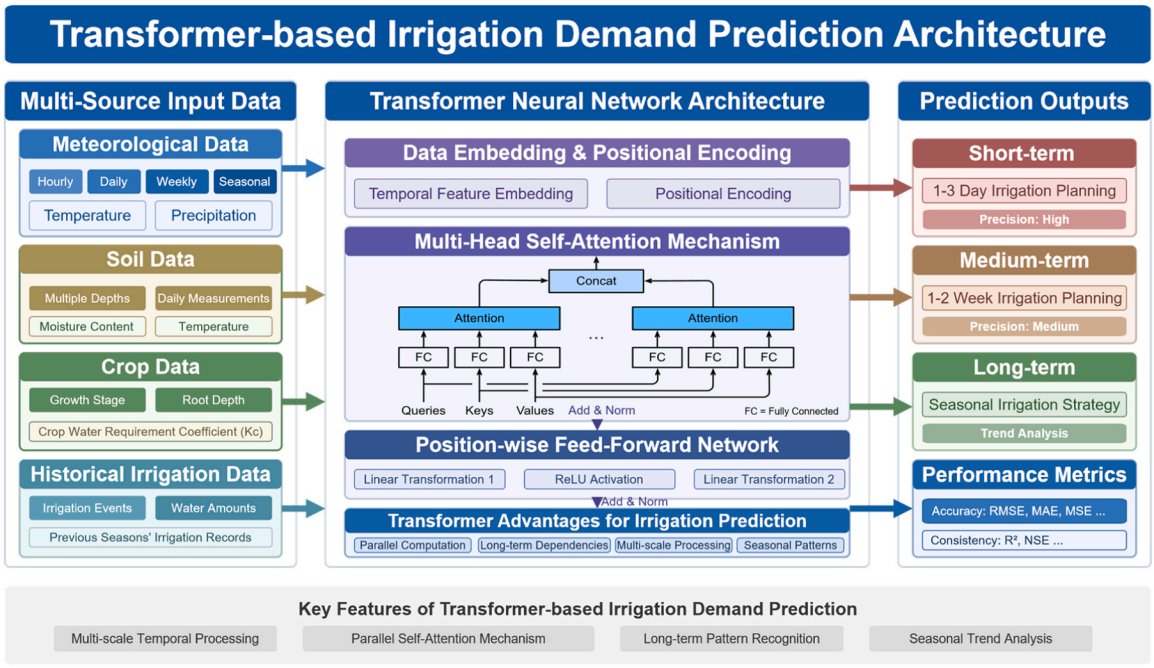


Fig. 4. Framework of a Transformer-based System for Multi-horizon Irrigation Demand Prediction.

Table 5
Performance of Hybrid DL Architectures in Predicting Key Irrigation-Related Variables.

Model	Reference	Prediction Object	Inputs	Performance	Theoretical Innovation
Transformer-GA	(González Perea et al., 2024)	Irrigation water demand (1-week ahead)	Historical irrigation data	99.9 % scenarios forecasted with 2.10 % error; 89.8 % accuracy improvement	Combined Transformer Neural Networks + fuzzy logic + Genetic Algorithms
CNN-GRU	(Yu et al., 2021)	SWC at different depths (maize root zone)	SWC + meteorological data	MSE < 1.0 for 10-day prediction; improved accuracy with soil depth	Integration of CNN (feature extraction) with GRU (memory capacity)
Physics-ConvLSTM	(Li et al., 2024)	SWC (1–16 days ahead)	Global Forecast System + in-situ station data	Outperformed pure DL models at 79.5 % of stations (16-day prediction)	Hybrid physics-DL model with attention mechanism & ensemble approach
CEEMDAN-CNN-GRU	(Ahmed et al., 2021)	SWC (1–30 days ahead)	Satellite (Global Land Data Assimilation System) + ground data (SILO) + climate indices (52 predictors)	Lower MAPE and RMSE than standalone/hybrid alternatives	Signal decomposition + CNN-GRU for multi-step forecasting
GRU-Transformer	(Zheng et al., 2024)	SWC (1–5 days ahead)	Meteorological + SWC time series	3-day ahead: MSE = 0.59 %, R^2 = 98.86 %	Integration of GRU with Transformer architecture
ResBiLSTM	(Yu et al., 2020)	SWC at multiple depths (20–50 cm, 1–6 days ahead)	Meteorological + SWC data + growth stage records	R^2 = [0.818–0.991], MAE = [0.79–2.00 %], MSE = [3.91–15.82 %]	Integration of ResNet (spatial) with BiLSTM (temporal) + meta-learner
BiLSTM-CNN-Attention	(Yan et al., 2024)	Irrigation volume for crops	Historical crop data+ meteorological data	RMSE = 0.005968, MAE = 0.004599, R^2 = 0.9749	BiLSTM (temporal) + CNN (spatial) + attention mechanism
LSTM with residual learning	(Li et al., 2022)	SWC (1–10 days ahead)	FLUXNET site data	7.95–19.71 % improvement over LSTM baseline	Encoder-decoder architecture with residual learning

modular division of labor, simultaneously processing spatial heterogeneity and temporal dynamics of irrigation data. For instance, the CNN-GRU model, using CNN for spatial feature extraction from meteorological data and GRU for temporal evolution, outperformed single models in SWC prediction (Yu et al., 2021). The ResBiLSTM architecture (Fig. 5) further advances multi-depth SWC prediction by combining ResNet for high-dimensional spatial feature extraction with BiLSTM for temporal sequence modeling via a meta-learner. This approach achieves high accuracy (R^2 = 0.818–0.991) across multiple soil depths (20–50 cm) and demonstrates varying performance across crop growth stages (Yu et al., 2020). Similarly, the GRU-Transformer architecture integrates GRU's short-term memory with Transformer's ability to capture long-distance dependencies, achieving an average R2 of 89.92 % in long-term prediction scenarios (Zheng et al., 2024). These studies collectively demonstrate how hybrid architectures enhance performance through specialized division of labor, making them particularly suitable for processing complex spatiotemporal data patterns in irrigation decisions.

Physics-integrated approaches combine mechanistic understanding

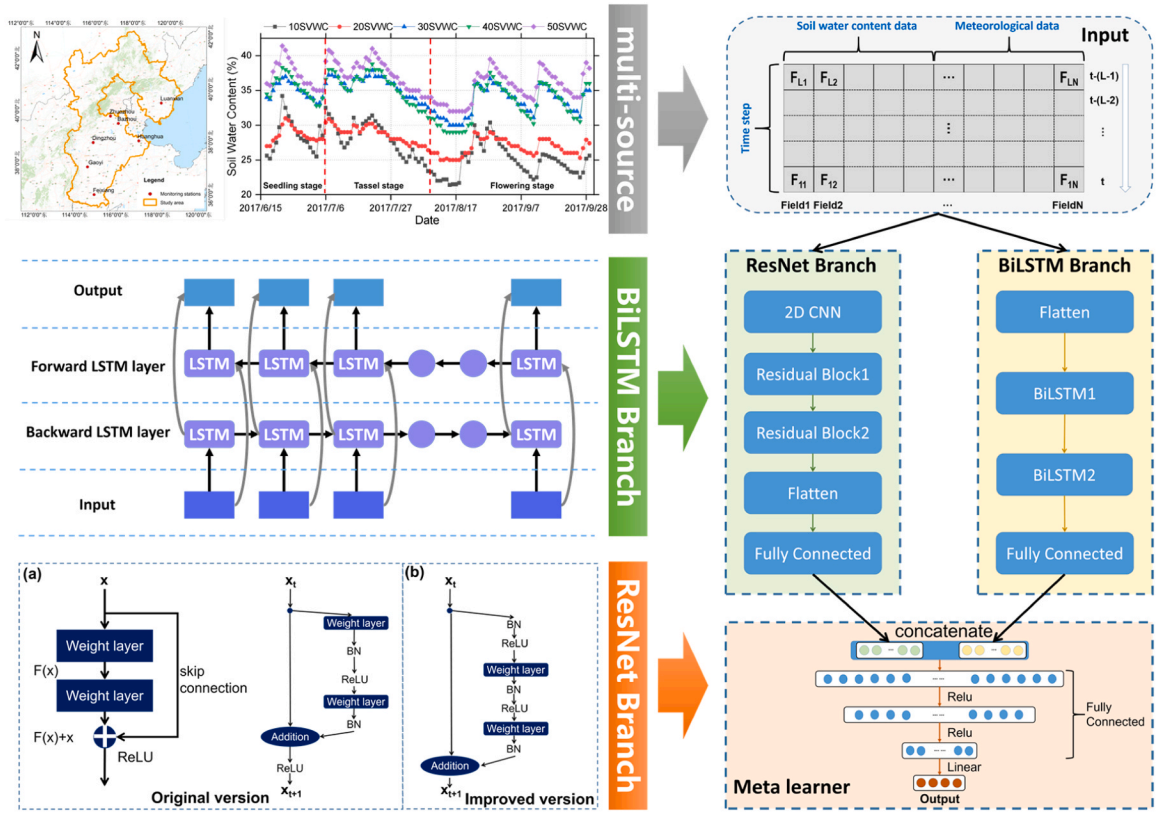


Fig. 5. The ResBiLSTM Hybrid Architecture for Multi-depth SWC Prediction (Yu et al., 2020).

with data-driven flexibility, ensuring physical consistency while providing computational efficiency for real-time decisions. For instance, combining physics-based modeling with DL approaches like ConvLSTM demonstrates complementary strengths in SWC prediction (Li et al., 2024).

The suitability of different architectures depends on application scenarios and resource constraints. Complex hybrid architectures are effective in data-rich environments (Yu et al., 2021), whereas data-limited scenarios benefit from pre-trained models (Islam and Yamane, 2021). For resource-constrained settings, lightweight systems are necessary (Rodić et al., 2022). This differentiated approach is key to the effective application of DL technologies in diverse irrigation environments.

4.5. Discussion

Temporal and hybrid models perform well under typical conditions, but their reliability often fails during the extreme weather events where they are most needed. This exposes a fundamental flaw in current evaluation methods, which prioritize average accuracy. Such an approach is inadequate for the high-stakes, risk-sensitive decisions required in agriculture. In contrast, intelligent irrigation systems' most critical application scenarios occur during abnormal conditions such as drought, where model performance often deteriorates substantially. Under extreme drought conditions, even leading hybrid models experience increased prediction errors, severely compromising decision reliability (Kelly et al., 2024). This limitation is compounded by the asymmetric error costs of irrigation, where crop losses from under-irrigation are far more severe than resource waste from over-irrigation.

To address this, the evaluation framework must shift from average accuracy to a comprehensive assessment of model robustness, adaptability, and uncertainty quantification. Frontier methods can achieve this by quantifying uncertainty. For instance, Bayesian DL provides posterior distributions rather than point estimates, deep ensemble methods express predictive uncertainty through inter-model differences, and probabilistic DL frameworks automatically adjust prediction intervals under abnormal conditions. These approaches hold significant advantages for irrigation prediction, particularly in reducing extreme errors and providing decision-makers with critical uncertainty information.

5. Deep reinforcement learning for irrigation decision-making

5.1. Framework for DRL in irrigation decision-making

Irrigation scheduling as a sequential decision problem requires balancing short-term benefits with long-term sustainability in a dynamic environment. DRL provides a suitable algorithmic framework for irrigation scheduling by combining the representational power of DNN with the decision optimization mechanism of RL. DRL framework for irrigation scheduling usually includes four key components (Fig. 6): state representation, action generation, reward calculation, and environment interaction.

State space design determines the DRL model's perception ability of the environment, typically including multi-dimensional information in irrigation decisions: soil parameters (e.g., SWC at different depths), meteorological data (temperature, rainfall), crop parameters (growth stage, root depth), and water resource status. The capacity for learning from unlabeled data makes DRL particularly suitable for data-scarce agricultural scenarios (Gautron et al., 2022). The action space defines the set of executable operations, which can be designed as discrete (e.g., gate control strategies) (Shahverdi et al., 2022) or continuous (e.g., fine irrigation volume adjustment) (Addas et al., 2024) based on system characteristics, requiring a balance between control precision and learning complexity.

Designing the reward function is a core challenge in DRL applications because it must quantify multiple conflicting objectives. The reward function for irrigation decisions is typically expressed as:

$$R = w_1 \cdot Y - w_2 \cdot WC - w_3 \cdot EC \tag{6}$$

where Y represents crop yield, WC represents water resource consumption, EC represents energy cost, and w_1, w_2, w_3 are the corresponding weights.

The primary optimization challenge is balancing immediate rewards with delayed returns, such as final yield, though well-designed reward functions can achieve dual optimization of resource efficiency and yield (Addas et al., 2024; Yang et al., 2020). The environment model, with which the agent interacts, requires a balance between simulation accuracy and computational efficiency, established crop models like DSSAT or AquaCrop are often used to create these learning environments (Balderas et al., 2025; Kelly et al., 2024).

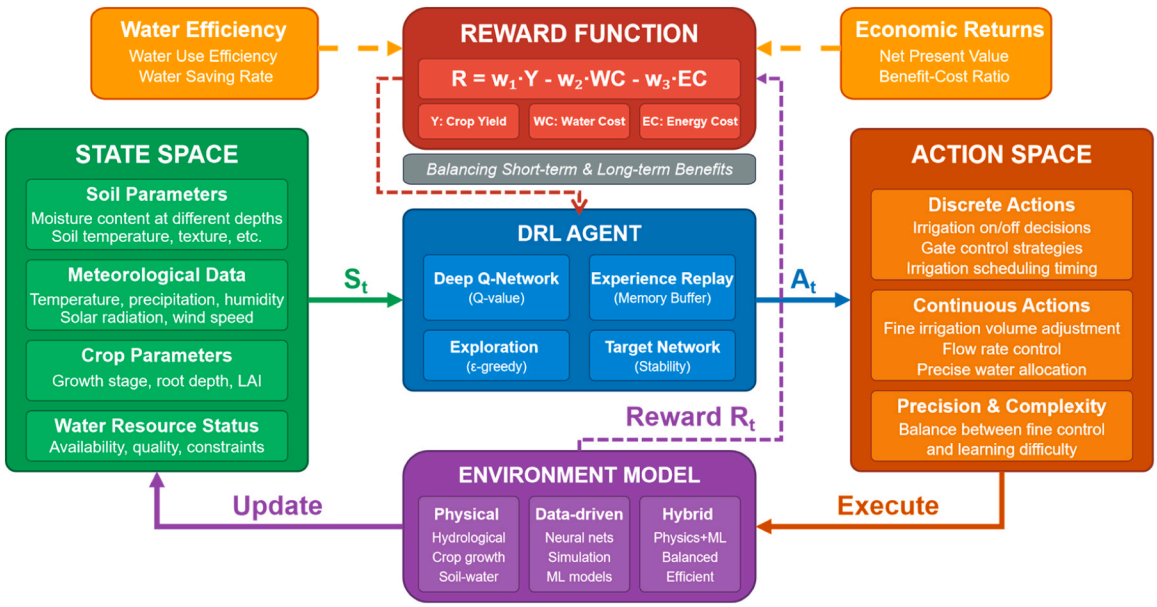


Fig. 6. Deep Reinforcement Learning Framework for Irrigation Scheduling Decisions.

5.2. Irrigation decision-making based on deep Q-learning

DQN maps environmental states to optimal irrigation actions by approximating the Q-value function with NNs. Improved variants such as Double DQN and Dueling architectures enhance decision stability and enable more refined water resource allocation.

DQN has demonstrated significant advantages across diverse irrigation scenarios. It has proven effective in hydraulic engineering, achieving high-precision control in canal systems (Shahverdi et al., 2022; Suntaranont et al., 2020), and at the field level, it optimizes resource allocation in AI-driven farms to enhance yield while reducing consumption (Addas et al., 2024). Theoretically, DRL is superior for handling the high-dimensional state spaces and long-term dependencies inherent in irrigation decisions (Yang et al., 2020).

As these applications illustrate in Table 6, DQN's core strengths include its ability to process complex, high-dimensional inputs (soil, meteorological, crop data), adapt to nonlinear system dynamics, balance short- and long-term objectives, and continuously improve strategies through autonomous learning. This makes DQN highly suitable for high-dimensional, sequential irrigation control.

5.3. Predictive irrigation decision-making with weather forecasts

Predictive irrigation decisions transform traditional reactive decisions into proactive decisions by incorporating weather forecast data, achieving significant improvements in WUE. The decision framework (Fig. 7) incorporates weather forecasts as an extended component of the DRL state space, enabling the model to develop anticipatory strategies based on predicted information. Chen et al. (2021) demonstrated that integrating multi-source forecast data reduces bias, with short-term forecasts (1–3 days) proving most valuable for immediate decisions and medium-term forecasts (4–7 days) supporting long-term planning. Field experiments in Nanchang rice cultivation showed significant benefits from this approach, with DRL systems saving 23 mm of water, reducing drainage by 21 mm, and decreasing irrigation frequency by 1.0 events per season. Beyond standalone DRL implementations, crop model integration represents another promising direction. Studies combining DSSAT crop models with DRL demonstrate enhanced decision optimization for cotton irrigation in water-scarce regions (Chen et al., 2024). Unlike traditional Model Predictive Control (MPC) methods, these integrated approaches do not require precise system models, learning directly from historical data and adapting to complex field environments while balancing water conservation with yield optimization.

5.4. Multi-objective irrigation optimization

Intelligent irrigation decision-making is essentially a multi-objective optimization problem, requiring balancing yield, WUE, and economic returns under resource constraints. DRL can effectively address this complex optimization problem through specific network structures and training strategies. One common method is the weighted sum, which linearly combines multiple objectives into a single reward signal, although setting the weights often relies on expert experience. For constraint handling, DRL frameworks mainly adopt penalty-based reward functions or constraint-aware action spaces. A study incorporates resource use penalties directly into the calculation of net return, which guides the agent towards optimized decisions under resource limitations (Yang et al., 2020). The AI-driven robotic farm is a typical case of multi-objective constrained optimization, where the system learned to improve resource utilization efficiency while satisfying crop growth needs, as evidenced by simultaneous increases in yield and reductions in water and chemical use (Addas et al., 2024).

Multi-objective constrained optimization faces challenges like managing conflicts between objectives and dynamically handling constraints. Advanced DRL frameworks can be designed to learn robust strategies. By carefully designing the reward function to penalize undesired outcomes, DRL models can be guided to find policies that not only maximize primary objectives like yield but also adhere to critical constraints, thereby improving learning efficiency and decision quality.

5.5. Discussion

DRL offers a powerful paradigm shift, replacing the static thresholds of traditional methods with adaptive policies learned through direct environmental interaction. This makes it effective for managing the time-lag effects, long-term dependencies, and nonlinear responses inherent in irrigation systems.

However, the significant gap between a model's performance in simulation and its effectiveness in the real world remains the greatest obstacle to the practical deployment of DRL in agriculture. This gap arises because physical models cannot fully capture environmental complexities, and acquiring sufficient real-world interaction data is often impractical (Umutoni and Samadi, 2024). Offline RL offers a viable solution, enabling models to learn effective policies from historical irrigation records without costly environmental interactions, thus improving sample efficiency. The value of DRL is particularly pronounced in water-stressed environments (Kelly et al., 2024). This highlights DRL's potential as an effective method for regions facing significant physical or regulatory water scarcity.

Despite theoretical breakthroughs, large-scale DRL application in irrigation decisions still lacks standardized simulation environments and benchmarks, with different studies employing different models and evaluation metrics, complicating fair algorithm performance comparison (Gautron et al., 2022). Future research must establish open irrigation simulation platforms and benchmark datasets, as seen with aquacrop-gym (Kelly et al., 2024), develop explainable DRL methods, and design hybrid intelligent systems capable of integrating human experience with automated decisions, to facilitate the transition from algorithmic theory to field practice.

Table 6
Comparative Analysis of DQN Applications Across Diverse Irrigation Scenarios.

Scenario	State Design	Action Design	Reward Function	Performance Metrics	Contributions	Reference
Canal Gate Control	Water level, flow rate, water level deviation	Gate opening adjustment	Sum of squared deviations from the target water level	Maximum error 11.6 %, average error 10.5 %	Adaptive control, no need for a precise model	(Shahverdi et al., 2022)
Irrigation Canal Control	Water depth, flow rate, rainfall forecast	Gate control parameters	Water level control precision	SEP = 2.58 cm, MAPE = 7.38 %	High-precision control, strong adaptability	(Suntaranont et al., 2020)
Crop Farm Irrigation	SWC, leaf temperature, growth stage	Irrigation volume, irrigation time	Yield gain-resource cost	26 % yield increase, 41 % water saving, 33 % chemical reduction	Multi-objective optimization, resource high efficiency	(Addas et al., 2024)
Automatic Irrigation Scheduling	Soil-crop-atmosphere system state	Irrigation decision (yes/no, water volume)	Long-term economic net return	Simulated net return is higher than traditional methods	Ability to handle high-dimensional states, long-term planning capability	(Yang et al., 2020)

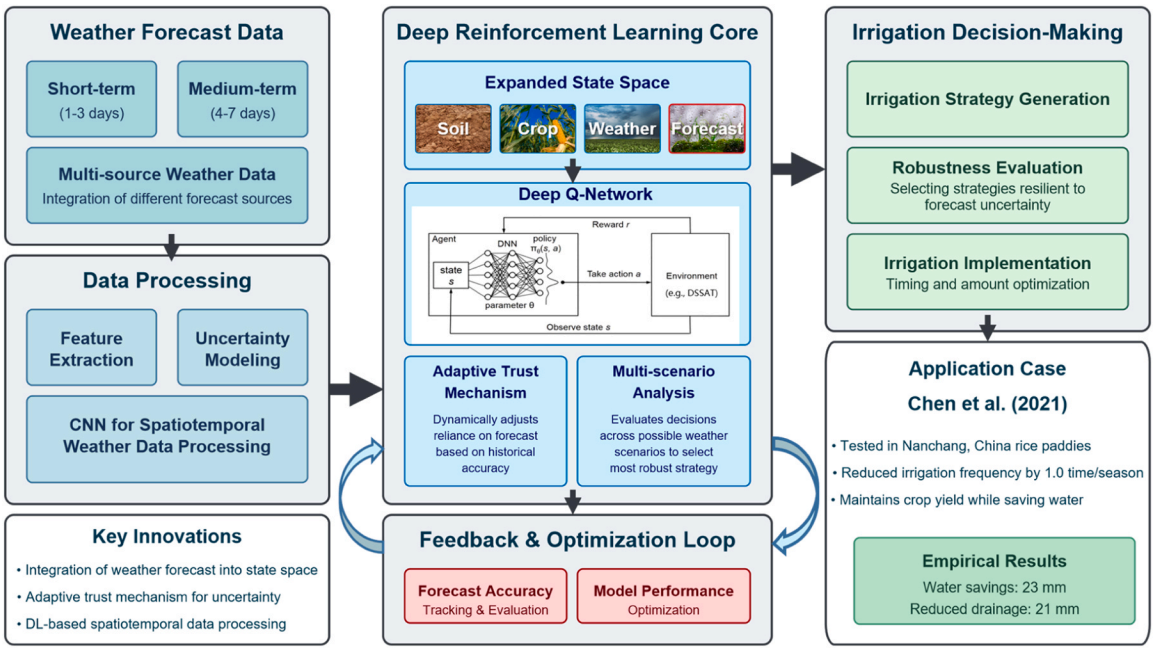


Fig. 7. A DRL-based Framework for Predictive Irrigation Decisions Incorporating Weather Forecasts.

6. System architecture and implementation

6.1. Collaborative architecture of IoT and DL

Intelligent irrigation systems integrate DL with IoT technology, forming a closed-loop for perception, learning, and decision-making. The perception layer relies on innovative sensor networks, from multi-parameter monitoring platforms for smallholder farmers (Dahane et al., 2022) to low-power LoRa systems that enable sensorless SWC monitoring to reduce deployment costs and energy consumption (Rodić et al., 2022).

To process this data, the edge-cloud collaborative architecture is widely adopted. This framework deploys lightweight DL models on edge devices for immediate tasks like data preprocessing, anomaly detection, and emergency response, while the cloud handles complex model training and global optimization (Ooi et al., 2023). This division of labor allows for efficient, real-time functionality. For instance, edge AI systems can use event-driven communication to send alerts only when anomalies are detected (Loukatos et al., 2022), while others use edge intelligence to distinguish between sensor failures and actual water stress, enhancing system reliability and user trust (Sabrina et al., 2022). Multi-layer architecture optimize data flow and computational resource allocation, ensuring the system can provide efficient, real-time irrigation decision support even under resource constraints in rural environments (Fig. 8).

6.2. Architectural design of real-time decision systems

Real-time decision systems must be optimized for the agriculture context, where the core challenge is deploying DL models under

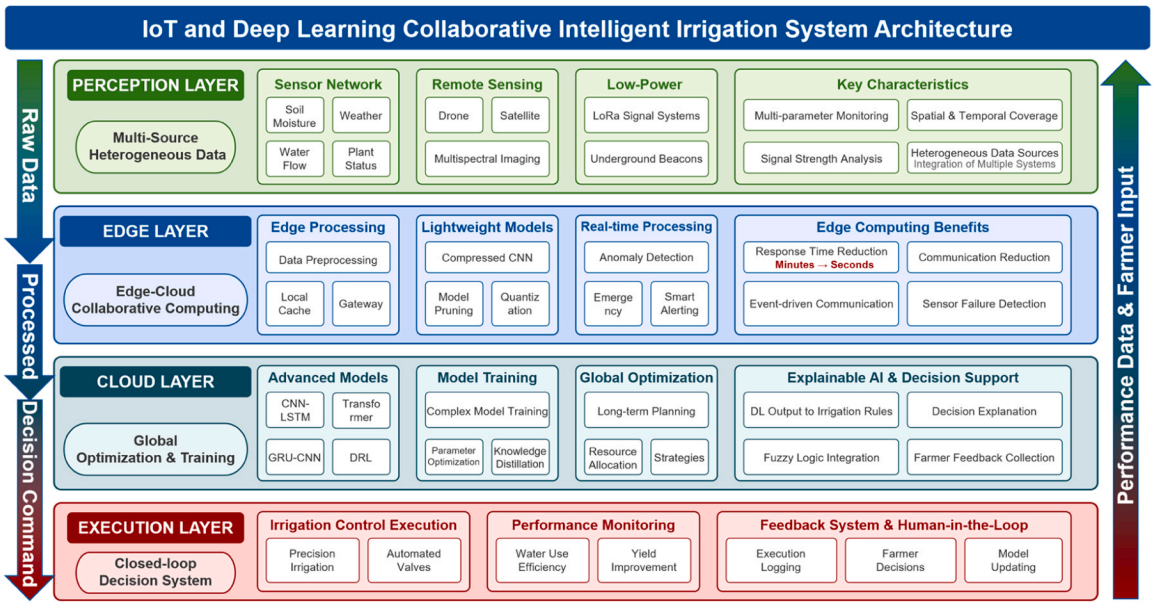


Fig. 8. Collaborative Intelligent Irrigation System Architecture of IoT and DL.

resource-constrained conditions. Lightweight design relies on dimensionality reduction, architecture optimization, and knowledge distillation. Nagappan et al. (2020) reduced inputs for the ETo prediction model from six variables to three through dimensionality reduction techniques. The HortNet417v1, developed by Islam and Yamane (2021), achieved 93 % classification accuracy for peach tree water stress. Knowledge distillation techniques can reduce model memory occupation by >70 % while maintaining performance. These technologies address the limitations in computational resources and energy consumption inherent to agricultural edge devices.

System architecture selection requires a trade-off between performance and resources, ranging from pure cloud to pure edge and hybrid architectures (Table 7). The edge-cloud collaborative architecture achieves the best balance through intelligent task allocation. The DLiSA system by Ooi et al. (2023) uses edge-side LSTMs for SWC prediction with an average response time of 60 ms and periodically synchronizes parameters with the cloud. Fault tolerance is equally important. Adaptive mechanisms are crucial for reliable system operation in harsh agricultural environments, demonstrating that DL not only improves decision accuracy but also enhances overall system robustness.

6.3. Farmer-oriented decision support systems

Farmer-oriented interfaces require explainability and usability. Explainable AI technologies address the "black box" nature of DL. Sabrina et al. (2022) uses fuzzy logic to transform the complex outputs of DNN into understandable irrigation rules, enabling farmers to understand the system's decision logic. The system simplifies complex multi-parameter DL models into intuitive rules, greatly increasing farmer acceptance. Furthermore, explanation modules display key environmental factors and their weights influencing current irrigation decisions, allowing farmers to understand the decision-making process of the DL models.

The system proposed by (Bhoi et al., 2021) pays special attention to feedback mechanism design, automatically adjusting DL models by collecting farmer feedback on irrigation decisions, creating a "**human-in-the-loop**" optimization system. The system records the actual effects of each irrigation operation, including yield indicators and WUE, as a basis for model adjustment. Through the closed-loop design, the system can adapt to local conditions and gradually improve decision accuracy while enhancing farmer trust in AI decisions.

6.4. System evaluation and practical application benefits

DL-enabled systems can save water, maintain or raise yields, and reduce inputs. Modern intelligent irrigation systems can achieve substantial water savings while increasing or maintaining crop yields (Table 8). The AI-driven robotic farm developed by Addas et al. (2024) utilized DRL to reduce water consumption by 41 %, increase crop yield by 26 %, and decrease chemical use by 33 %. Similarly, RL strategies have proven effective in specific crops. For cotton, one system improved yield while simultaneously reducing water consumption (Chen et al., 2024), and another for rice saved an average of 23 mm of irrigation water and reduced drainage by 21 mm without significant yield loss (Chen et al., 2021). Despite potentially higher initial investments, the savings on water, energy, and labor can lead to favorable recovery periods. A system reduced overall cultivation time by 40.94 % and lowered energy consumption for data transmission by 97.8 % (Ma et al., 2022). Furthermore, a hybrid Transformer network for precision irrigation demonstrated direct cost savings of 10.3 % alongside water savings of 10.8 % (Madhukumar et al., 2024).

User acceptance, a critical factor for adoption, is strongly linked to model interpretability. Systems that are transparent and easily understood by farmers are more likely to be trusted and utilized (Sabrina et al., 2022). Market trends reflect this sentiment, Martey et al. (2023) found that 98 % of surveyed farmers expressed a willingness to pay for precision irrigation information, with higher interest among women, youth, and high-income groups. This economic willingness validates the perceived benefits and provides a market foundation for commercialization. Ultimately, DL-driven intelligent irrigation systems, especially in regions facing significant physical or regulatory water scarcity, present an economically feasible and vital technical path towards sustainable agriculture (Kelly et al., 2024).

Table 8
Performance and Benefits of DL-Driven Intelligent Irrigation Systems in Practical Applications.

System	Reference	Application Scenario	Benefits and Performance
AI-driven Robotic Farm	(Addas et al., 2024)	Simulated robotic crop farm	41 % reduction in water utilization; 26 % increase in crop yield; 33 % reduction in chemical use
DRL with Weather Forecast	(Chen et al., 2021)	Paddy rice in Nanchang, China	23 mm average savings in irrigation water per season; 21 mm reduction in drainage; Achieved water conservation without significant yield reduction
Smart Agricultural Planting System	(Ma et al., 2022)	Intelligent farm management system	50 % & 30 % reduction in water consumption; 97.8 % reduction in communication transmissions; 40.94 % reduction in cultivation time
Hybrid Transformer Network	(Madhukumar et al., 2024)	Precision irrigation at test sites	10.85 % reduction in water usage through optimized irrigation schedules; 10.34 % direct cost savings
Wetting Front Detection System	(Martey et al., 2023)	Survey of smallholder farmers	98 % of surveyed farmers expressed willingness to pay for the tool; Information on the tool increased willingness

Table 7
Comparison of System Architectures for Real-time Irrigation Decision-Making.

System Architecture	Computational Complexity	Deployment Cost	Reliability	Scalability	Scenarios	Reference
Pure Cloud Computing	Supports complex models	Low	Medium (Network dependent)	High	Model training, large-scale analysis	(Ooi et al., 2023)
Pure Edge Computing	Supports lightweight models	High	High (Network independent)	Medium	Low-latency, offline tasks	(Ooi et al., 2023)
Edge-Cloud Collaboration	Balanced: Edge (light) & Cloud (complex)	Medium	High (with redundancy)	High	Balanced real-time & complex tasks	(Loukatos et al., 2022)
Multi-level Architecture	Dynamic task-adaptive allocation	Medium-high	High (hierarchical tolerance)	High	Large-scale, heterogeneous environments	(Man et al., 2024)

6.5. Discussion

The "black-box" nature of DL models conflicts with farmers' experience-based decision-making, creating a trust deficit that hinders adoption. To narrow this gap, explainability methods are critical. Techniques like rule extraction and feature importance analysis can distill complex models into intuitive insights, enhancing farmer trust and integrating local agronomic knowledge.

Furthermore, static "train-then-deploy" systems struggle with dynamic agricultural environments. Continuous learning is essential for adaptation. Incremental learning frameworks can prevent model degradation over time (van de Ven et al., 2022; Yang et al., 2020), while human-machine collaborative frameworks that integrate farmer feedback can enhance decision robustness. For long-term sustainability, multi-level architectures and community sharing models offer cost-effective pathways to improve technology acceptance.

Ultimately, future intelligent irrigation systems should evolve into "socio-technical" complexes. This requires incorporating sociological and behavioral science perspectives to construct sustainable solutions that are both technologically advanced and socially embedded (Ara et al., 2021).

7. Challenges, opportunities, and future directions

7.1. Technical challenges

Addressing the key technical challenges of data acquisition, model generalization, and deployment requires integrated solutions (Fig. 9).

Data acquisition is hindered by the high spatiotemporal variability of agricultural environments and by infrastructure and digital literacy gaps in rural areas (Arif et al., 2024; Richards et al., 2023). Multi-source data fusion, which integrates satellite, UAV, and ground sensor data, offers a robust solution. While, model generalization is a critical challenge, as models trained in one setting often fail in new environments with diverse soils, climates, and crops, a problem intensified by climate change which demands more adaptive models (Palutikof et al., 2019). Domain adaptation and transfer learning are effective strategies for transferring models from data-rich to data-scarce regions. Furthermore, practical deployment is constrained by limited computational resources in rural IoT settings, which conflicts with the high demands of DL models, particularly for smallholder farmers (Dahane et al., 2022; Ooi et al., 2023). While edge computing provides a partial solution, model optimization remains crucial. Key techniques include model compression, quantization, knowledge distillation, and the design of lightweight architectures or reduced-feature models (Islam and Yamane, 2021; Jiang et al., 2021; Nagappan et al., 2020; Tian et al., 2025; Xiao et al., 2024).

7.2. Application challenges

Applying DL in irrigation faces both general AI challenges and specific agricultural barriers. Table 9 summarizes the main barriers to intelligent irrigation system adoption and resolution strategies.

Cost and acceptability are major barriers. Clay et al. (2024) point out that adoption barriers include costs and farmers' unwillingness, which pose challenges for DL, which typically requires more data and computation. To mitigate these hardware and resource constraints, strategies like shifting computation from centralized cloud servers to edge devices closer to the data source can be employed, reducing network latency and reliance on powerful, expensive infrastructure (Ooi et al., 2023). Solutions for smallholder farmers include community sharing models (where multiple farmers share one system to spread costs) and tiered functionality design (basic functions provided at low cost, advanced functions paid for as needed). The explainable AI system proposed by Sabrina et al. (2022) increases trust by improving decision transparency, addressing the core issue of farmer reluctance or distrust.

User skill requirements and regional adaptability are significant barriers to the practical application of DL systems. Maintaining DL models requires professional knowledge that most farmers lack, but this threshold can be lowered. For instance, explainable AI can transform complex model outputs into intuitive rules (Sabrina et al., 2022), while tiered interfaces and community training can enhance usability and technology acceptance. A related challenge is regional adaptation, as models trained in one location are difficult to apply directly across diverse agricultural environments. Therefore, assessing a model's generalizability is crucial (Zhang et al., 2020) with solutions including fine-tuning pre-trained models with local data or developing DL architectures with built-in adaptation mechanisms.

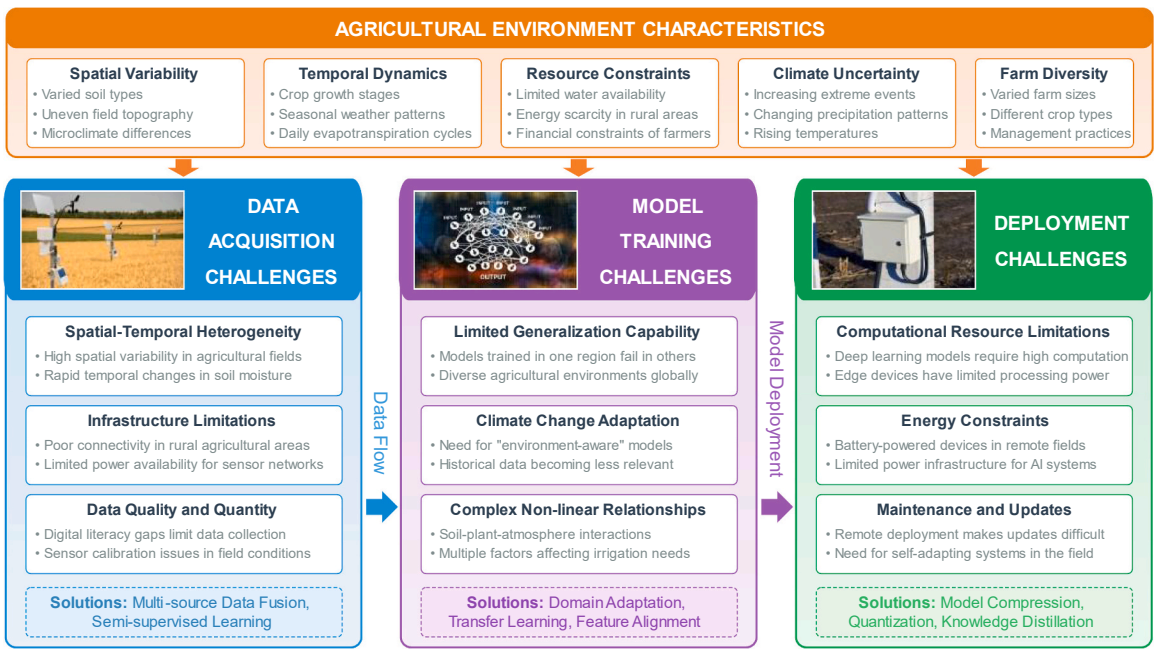


Fig. 9. Technical Challenges and Solutions Framework for DL in Intelligent Irrigation.

Table 9
Main Barriers to Intelligent Irrigation System Adoption and Resolution Strategies.

Barrier Type	Specific Manifestation	Impact Level	Resolution Strategy	Case Study
Cost Factors	High hardware requirements for DL	High	Cloud training + edge inference	Edge-cloud collaborative systems (Ooi et al., 2023)
Technical Complexity	DL models are difficult to understand	Medium-high	Explainable AI framework	Decision rule visualization (Sabrina et al., 2022)
Farmer Distrust	Doubt about AI "black box" decisions	High	Demand-driven participatory approaches and user training	Co-design workshops with end-users (Ara et al., 2021)
Regional Adaptability	Difficulty in model transfer	High	Model transferability assessments and fine-tuning	Cross-regional transferability assessment of (Zhang et al., 2020)
Digital Divide	Insufficient AI literacy among farmers	Medium-high	User-centric design and improved usability	Simplified interfaces, closing digital literacy gaps (Richards et al., 2023)
System Adaptability	Static models fail to adapt to changing environments	High	Implement continuous or incremental learning frameworks	Adaptive frameworks with parameter regularization (van de Ven et al., 2022)

7.3. Future research directions in the context of climate change

Climate change motivates three key research directions for DL in intelligent irrigation. Climate-adaptive decision models are needed to challenge the assumption of climate stability amid predictions of increased drought and severe seasonal water scarcity (Essa et al., 2023; McNamara et al., 2021). Climate-crop-water coupling models represent a key technology to balance short-term irrigation demands with long-term water sustainability. While, extreme weather response technologies must shift from passive reaction to proactive prevention, using AI for early warning against intensified precipitation and flooding (Singh and Goyal, 2023; Tabari, 2020). Furthermore, ultra-precise management in water-scarce regions will require integrating innovations like comprehensive optimization strategies and wastewater reuse to address the global water crisis, providing ideal scenarios for DL applications (Christou et al., 2024; Dinar et al., 2019; Mekonnen and Hoekstra, 2016; Pham et al., 2021). These interconnected research paths form an intelligent closed-loop system—from prediction and decision to implementation—to jointly address the complex challenges brought by climate change.

7.4. Potential of emerging DL technologies in irrigation decision-making

Emerging DL methods offer novel solutions for the core challenges of data limitation, computational constraints, and environmental adaptability in irrigation. These technologies can be integrated through comprehensive data fusion frameworks, forming a solution chain from data acquisition to decision optimization (Man et al., 2024).

Federated and self-supervised learning address data bottlenecks by enabling collaborative training while protecting privacy and extracting features from unlabeled data (Saeed et al., 2021). To manage limited computational resources, Neural Architecture Search (NAS) automates lightweight model design (Younesi et al., 2024). Environmental adaptability is enhanced through few-shot learning, which allows models to adapt to new irrigation areas with minimal samples, and causal inference, which identifies intervention relationships for better decision precision (Ahmed et al., 2023; Wu et al., 2025).

Large language models (LLMs) and multimodal AI represent transformative technologies that can reshape irrigation decision support systems. LLMs offer value by integrating unstructured agricultural information and providing intuitive natural language interfaces (Sun et al., 2024; Tzachor et al., 2023). Multimodal AI integrates visual, textual, and numerical data from diverse sources like UAVs to diagnose water stress and convert complex data into farmer-oriented narrative recommendations (El Sakka et al., 2025; Zhang et al., 2023). At the watershed level, multi-agent reinforcement learning provides a collaborative decision-making framework to improve water use efficiency.

Future research should focus on technology integration and practical deployment. This includes developing domain-specific small language models, building knowledge graph-enhanced hybrid systems, and creating human-AI collaborative frameworks that integrate farmer expertise with AI capabilities to achieve sustainable water resource management under climate change (Sun et al., 2024).

8. Conclusion

DL technologies have transformed intelligent irrigation decision-making systems, offering precise solutions to the complex challenges of water resource management in agriculture. Based on our comprehensive review, we draw the following key conclusions:

(1) DL approaches, by processing complex data patterns, have reframed irrigation decision-making. Traditional irrigation methods typically rely on simple thresholds or expert rules, failing to capture nonlinear relationships between environmental factors and water requirements. DL has fundamentally reframed irrigation as a data-driven prediction and optimization problem, rather than a reactive, experience-based practice. By automatically modeling complex, nonlinear relationships between environmental variables and crop water needs, these technologies enable a paradigm shift towards proactive and precise water management.

(2) Each DL architecture demonstrates distinct comparative advantages in different irrigation decision components. CNN-based models excel in irrigation system mapping with up to 99 % precision, extracting critical spatial features from RS data. RNN variants achieve high accuracy in SWC dynamics prediction (R^2=0.818–0.991) by capturing temporal patterns crucial for irrigation timing decisions. Transformer architecture delivers up to 99.9 % accurate long-term irrigation forecasting through the ability to model extended dependencies across growing seasons. DRL balances multiple competing objectives in irrigation scheduling, simultaneously achieving 41 % water savings and 26 % yield increases.

(3) Hybrid models that integrate multiple architectures and knowledge sources consistently outperform single-approach systems in real-world conditions. The CNN-GRU hybrid architecture improves SWC prediction accuracy by simultaneously processing spatial heterogeneity and temporal dynamics. Physics-guided DNNs incorporating domain knowledge demonstrate higher robustness under extreme climate conditions, addressing a critical limitation of purely data-driven approaches. The hybrid systems are particularly valuable in agricultural environments characterized by high variability and limited data availability.

(4) Edge-cloud collaborative architecture represents the most effective framework for deploying DL models in agricultural environments. By distributing computational tasks based on urgency and resource requirements, the systems achieve bandwidth reduction and decrease anomaly response time from minutes to seconds. Edge intelligence enables critical real-time processing despite rural connectivity limitations, while cloud infrastructure supports complex model training and optimization. This architectural approach effectively bridges the gap between sophisticated algorithms and practical field implementation.

Future research should focus on developing end-to-end, closed-loop

systems that unify the currently separate "prediction-decision" frameworks. Three key directions deserve particular attention: (1) climate-adaptive decision models capable of handling increasing uncertainties in precipitation patterns and extreme events; (2) uncertainty-aware frameworks that provide probabilistic predictions rather than point estimates, enabling comprehensive risk assessment under variable conditions; and (3) integration of emerging technologies such as federated learning and self-supervised learning to overcome data limitations while ensuring privacy and improving model robustness across diverse agricultural environments.

CRediT authorship contribution statement

Qinglin Qu: Writing – review & editing, Visualization, Validation, Resources, Project administration. **Jingxin Yu:** Writing – review & editing, Writing – original draft, Validation, Funding acquisition, Formal analysis, Conceptualization. **Congcong Sun:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology. **Yinkun Li:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Data curation. **Xiaoming Wei:** Writing – original draft, Validation, Supervision, Resources, Data curation. **Shuyi Peng:** Writing – review & editing, Writing – original draft, Visualization, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Thanks to the support from the Youth Fund of Beijing Academy of Agriculture and Forestry Sciences, China (QNJJ202410), the Reform and Development Project of Beijing Academy of Agriculture and Forestry Sciences, China (GGFZ20240107), and the China Scholarship Council (No.202409110090).

Data availability

No data was used for the research described in the article.

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